COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	05/10/2026	**DATE SUBMITTED:**	06/10/2026
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ACTIVITY NO. 10. 1 **Install, Configure and Manage Enterprise Log Monitoring via Ansible**

1. PROCEDURE OUTPUT

1. Create a playbook that:
    a. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash)
2. Apply the concept of creating roles.
3. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
4. Show an output of the installed Elastic Stack for both Ubuntu and CentOS.
5. Make sure to create a new repository in GitHub for this activity.

Output:

1. **Directories and Files**

```
erickalvarez@alvarezworkstation:~/ACT10$ tree
.
├── ansible.cfg
├── inventory.ini
├── README.md
└── roles
    ├── elasticsearch
    │   └── tasks
    │       └── main.yml
    ├── kibana
    │   └── tasks
    │       └── main.yml
    └── logstash
        └── tasks
            └── main.yml

7 directories, 6 files
erickalvarez@alvarezworkstation:~/ACT10$
```

tree

- This is the main structure of the repository.

```
erickalvarez@alvarezworkstation:~/ACT10$ cat inventory.ini
[elasticsearch]
192.168.56.105 ansible_user=alvarezworkstation

[kibana]
192.168.56.108 ansible_user=ealvarez

[Logstash]
192.168.56.107 ansible_user=alvarezworkstation


erickalvarez@alvarezworkstation:~/ACT10$
```

Inventory.ini

- This is the screenshot of what my inventory.ini looks like. I separated the three machines into elasticsearch, kibana, and logstash groups. This makes it easier for Ansible to know which machine will handle each part of the Elastic Stack.

```
erickalvarez@alvarezworkstation:~/ACT10$ cat ansible.cfg
[defaults]
inventory = inventory.ini
remote_user = alvarezworkstation
host_key_checking = False

[privilege_escalation]
become = True
become_method = sudo
become_ask_pass = True
erickalvarez@alvarezworkstation:~/ACT10$
```

Ansible.cfg

- Ansible.cfg is the configuration for the ansible to read. For example, in the default function, we set inventory.ini as our main inventory.

Step 2: Installation:

```
erickalvarez@alvarezworkstation:~/ACT10$ cat site.yml
---
- name: Install Elasticsearch
  hosts: all
  become: true
  roles:
    - role: elasticsearch
      when: inventory_hostname in groups['elasticsearch']

- name: Install Kibana
  hosts: all
  become: true
  roles:
    - role: kibana
      when: inventory_hostname in groups['kibana']

- name: Install Logstash
  hosts: all
  become: true
  roles:
    - role: logstash
      when: inventory_hostname in groups['logstash']

erickalvarez@alvarezworkstation:~/ACT10$
```

Site.yml

```
---
- name: Install required packages
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - curl
      - gnupg
    state: present
    update_cache: true

- name: Download Elastic GPG key
  ansible.builtin.get_url:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /tmp/elasticsearch.asc
    mode: "0644"

- name: Convert GPG key
  ansible.builtin.command:
    cmd: gpg --dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg /tmp/ela
sticsearch.asc
  args:
    creates: /usr/share/keyrings/elasticsearch-keyring.gpg

- name: Add Elastic repository
```

```
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg] https:/
/artifacts.elastic.co/packages/9.x/apt stable main"
    state: present

- name: Install Elasticsearch
  ansible.builtin.apt:
    name: elasticsearch
    state: present
    update_cache: true

- name: Configure Elasticsearch
  ansible.builtin.copy:
    dest: /etc/elasticsearch/elasticsearch.yml
    content: |
      cluster.name: elastic-lab
      node.name: elasticsearch
      network.host: 0.0.0.0
      http.port: 9200
      discovery.type: single-node
    owner: root
    group: elasticsearch
    mode: "0660"
```

```
    mode: '0660'

- name: Enable Elasticsearch
  ansible.builtin.systemd:
    name: elasticsearch
    enabled: true
    state: started
erickalvarez@alvarezworkstation:~/ACT10/roles/elasticsearch/tasks$ SSS
```

Elasticsearch main.yml

- Elasticsearch role installs and configures Elasticsearch on Ubuntu. First, Ansible installs all dependencies, imports Elastic GPG key, adds the Elastic repository and installs Elasticsearch, then creates all needed directories for configuration and logging, and finally deploys configuration file elasticsearch.yml. The last step of role is enabling and starting Elasticsearch service that would be able to accept and store data

```
erickalvarez@alvarezworkstation:~/ACT10/roles/kibana/tasks$ cat main.yml
---
- name: Install required packages
  ansible.builtin.dnf:
    name:
      - curl
      - gnupg2
    state: present

- name: Import Elastic GPG key
  ansible.builtin.rpm_key:
    key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    state: present

- name: Add Elastic repository
  ansible.builtin.yum_repository:
    name: elastic-9.x
    description: Elastic 9.x Repository
    baseurl: https://artifacts.elastic.co/packages/9.x/yum
    enabled: true
    gpgcheck: true
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
```

```
- name: Install Kibana
  ansible.builtin.dnf:
    name: kibana
    state: present

- name: Configure Kibana
  ansible.builtin.copy:
    dest: /etc/kibana/kibana.yml
    content: |
      server.host: "0.0.0.0"
      server.port: 5601
      elasticsearch.hosts:
        - "http://192.168.56.101:9200"
    owner: root
    group: kibana
    mode: "0660"

- name: Enable Kibana
  ansible.builtin.systemd:
    name: kibana
    enabled: true
    state: started
erickalvarez@alvarezworkstation:~/ACT10/roles/kibana/tasks$ SD
```

Kibana main.yml

- Kibana role installs and configures Kibana on CentOS. First, Ansible installs all needed packages, imports Elastic GPG key, adds Elastic repository and installs Kibana. The configuration file makes Kibana to listen on network and connect to Elasticsearch using its IP address and port number. The final step of role is enabling and starting Kibana service.

```
---
- name: Install required packages
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - curl
      - gnupg
    state: present
    update_cache: true

- name: Download Elastic GPG key
  ansible.builtin.get_url:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /tmp/logstash.asc
    mode: "0644"

- name: Convert GPG key
  ansible.builtin.command:
    cmd: gpg --dearmor -o /usr/share/keyrings/elastic-keyring.gpg /tmp/logstash.
asc
  args:
    creates: /usr/share/keyrings/elastic-keyring.gpg

- name: Add Elastic repository
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elastic-keyring.gpg] https://artif
acts.elastic.co/packages/9.x/apt stable main"
    state: present

- name: Install Logstash
  ansible.builtin.apt:
    name: logstash
    state: present
    update_cache: true

- name: Configure Logstash
  ansible.builtin.copy:
    dest: /etc/logstash/logstash.yml
    content: |
      path.data: /var/lib/logstash
      path.logs: /var/log/logstash
    owner: root
    group: logstash
    mode: "0644"

- name: Enable Logstash
  ansible.builtin.systemd:
    name: logstash
```

```
- name: Enable Logstash
  ansible.builtin.systemd:
    name: logstash
    enabled: true
    state: started
erickalvarez@alvarezworkstation:~/ACT10/roles/logstash/tasks$ S
```

Logstash main.yml

- The Logstash role takes care of the installation and configuration of the Logstash service on the Ubuntu server. The installation of the necessary dependencies, adding the Elastic repositories and installing the Logstash service is done by Ansible. The initial configuration file for Logstash regarding the directories of its data and logs is created. Lastly, the Logstash service is started and enabled.

```
PLAY RECAP *********************************************************************
192.168.56.105             : ok=7    changed=0    unreachable=0    failed=1    s
kipped=0    rescued=0    ignored=0
192.168.56.107             : ok=10   changed=3    unreachable=0    failed=0    s
kipped=13   rescued=0    ignored=0
192.168.56.108             : ok=9    changed=0    unreachable=0    failed=0    s
kipped=14   rescued=0    ignored=0

erickalvarez@alvarezworkstation:~/ACT10$
```

Execution

- I encountered multiple errors but this recap shows that the installation worked successfully. The only problem here was that the service of elasticsearch did not run automatically.

```
PLAY RECAP *********************************************************************
192.168.56.105             : ok=12   changed=7    unreachable=0    failed=0
kipped=13   rescued=0    ignored=0
192.168.56.107             : ok=10   changed=0    unreachable=0    failed=0
kipped=15   rescued=0    ignored=0
192.168.56.108             : ok=9    changed=0    unreachable=0    failed=0
kipped=16   rescued=0    ignored=0

erickalvarez@alvarezworkstation:~/ACT10$
```

After fixing the error

4. Output

Elasticsearch:

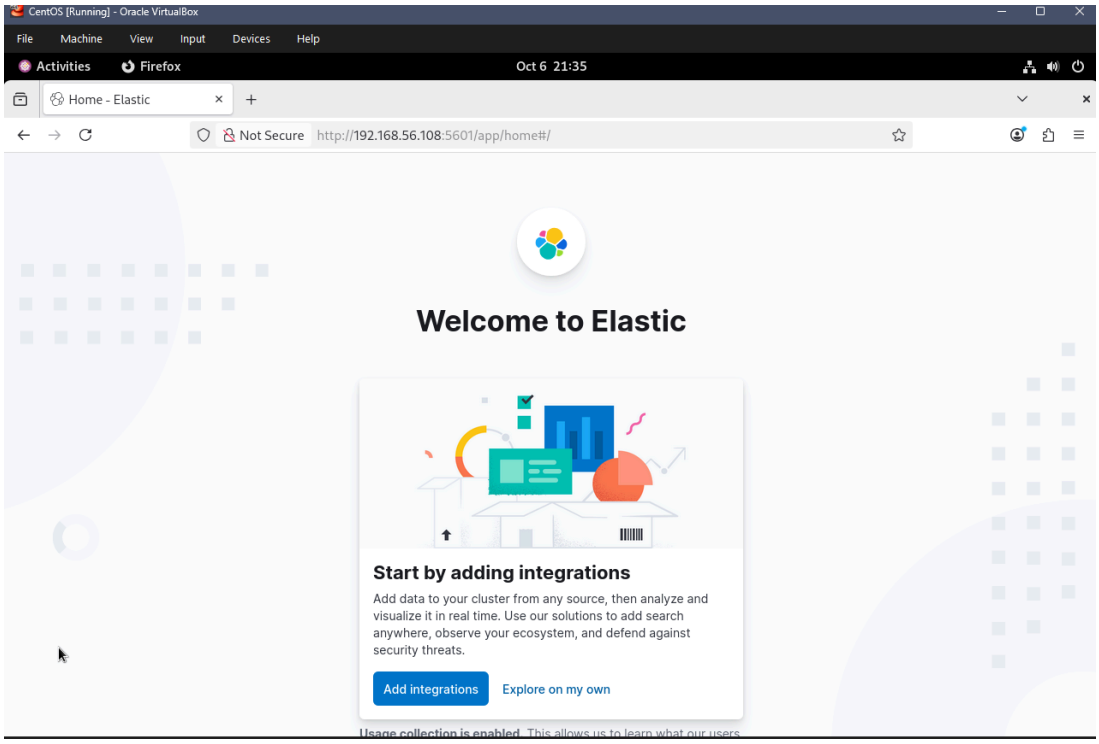
```
erickalvarez@alvarezworkstation:~/ACT10$ ssh erickalvarez@192.168.56.105
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-138-generic x86_64)

 * Documentation:  https://help.ubuntu.com
```

```
erickalvarez@alvarezworkstation:~$ curl http://localhost:9200
{
  "name" : "node-1",
  "cluster_name" : "my-application",
  "cluster_uuid" : "5k2LIbOyR8WaAfkN4ErEPQ",
  "version" : {
    "number" : "8.19.23",
    "build_flavor" : "default",
    "build_type" : "deb",
    "build_hash" : "6940e499f8a38841809b8e295a20c6421c95d9e4",
    "build_date" : "2026-09-30T23:11:55.493565832Z",
    "build_snapshot" : false,
    "lucene_version" : "9.12.2",
    "minimum_wire_compatibility_version" : "7.17.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "You Know, for Search"
}
```

- It can be seen that the installation of the Elasticsearch was successful, and it is running on the Ubuntu server. The systemctl status elasticsearch output shows that the service is active, and the curl http://localhost:9200 output verifies that Elasticsearch is able to answer. This indicates that the Elasticsearch server is functional and is able to save and process the data for the other components of Elastic Stack.

Kibana:



- It can be seen that the installation of Kibana was successful, and it is running on the CentOS server. The output shows the service status of the Kibana as running, and it can also be seen that accessing its web interface is successful. Moreover, Kibana has been set up such that it connects to the Elasticsearch server using its IP address and port.

Logstash:

```
*** System restart required ***
Last login: Tue Oct  6 20:53:53 2026 from 192.168.56.111
erickalvarez@alvarezworkstation:~$ sudo systemctl status logstash
[sudo] password for erickalvarez:
● logstash.service - logstash
     Loaded: loaded (/lib/systemd/system/logstash.service; enabled; vendor pres>
     Active: active (running) since Tue 2026-10-06 21:05:33 +08; 11s ago
   Main PID: 51435 (java)
      Tasks: 22 (limit: 4598)
     Memory: 522.1M
        CPU: 20.973s
     CGroup: /system.slice/logstash.service
             └─51435 /usr/share/logstash/jdk/bin/java -Xms1g -Xmx1g -Djava.awt.>

Oct 06 21:05:33 alvarezworkstation systemd[1]: Started logstash.
Oct 06 21:05:33 alvarezworkstation logstash[51435]: Using bundled JDK: /usr/sha>
```

- Shows that Logstash was installed and enabled in the Ubuntu server. The status of the service proves that the Logstash is running, while the command to check the version of the Logstash proves that the right software has been installed. Logstash is known for collecting, processing, and sending the log data.

Github:

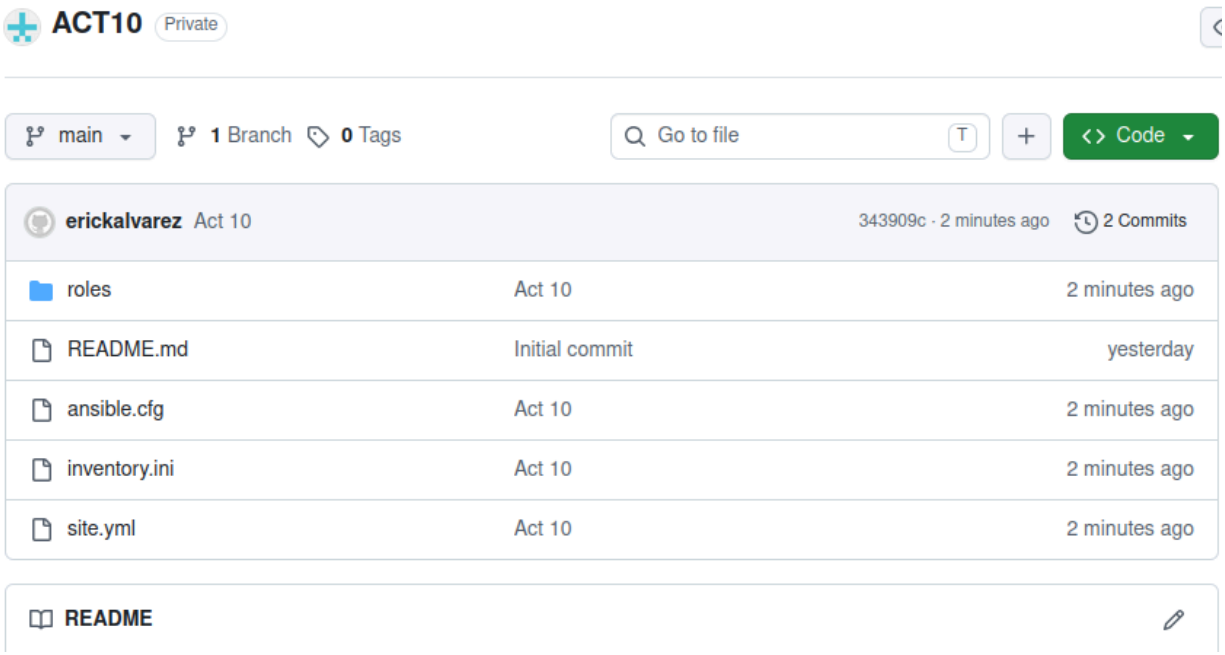
```
erickalvarez@alvarezworkstation:~/ACT10$ git add .
erickalvarez@alvarezworkstation:~/ACT10$ git commit -m "Act 10"
[main 343909c] Act 10
 Committer: erickalvarez <erickalvarez@alvarezworkstation.myguest.virtualbox.org>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 6 files changed, 206 insertions(+)
 create mode 100644 ansible.cfg
 create mode 100644 inventory.ini
 create mode 100644 roles/elasticsearch/tasks/main.yml
 create mode 100644 roles/kibana/tasks/main.yml
 create mode 100644 roles/logstash/tasks/main.yml
 create mode 100644 site.yml
erickalvarez@alvarezworkstation:~/ACT10$ git push origin main
Username for 'https://github.com': ErickAlvarezz
Password for 'https://ErickAlvarezz@github.com':
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 2 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (15/15), 2.66 KiB | 1.33 MiB/s, done.
Total 15 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/ErickAlvarezz/ACT10
   d84c678..343909c  main -> main
erickalvarez@alvarezworkstation:~/ACT10$ s
```



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Conclusion:
In conclusion, the Elastic Stack was successfully set up in the project, as Ansible was used to install it on three different virtual machines. Elasticsearch was installed in Ubuntu, Kibana in CentOS, and Logstash in another Ubuntu machine. The deployment was made easier by Ansible as it installed all the required packages, repositories, configurations, and services automatically.

However, there were some problems encountered during the installation process, namely, APT Signed-By error due to the presence of several Elastic repository entries with different GPG keys, Elasticsearch configuration directory error, Elasticsearch service execution error, and an Elasticsearch SSL configuration error due to leftover security settings. Another problem was the failure of Kibana at first since xpack.security.enabled couldn't be changed in the current version of it. Also, the important error was the mismatch between Kibana and Elasticsearch versions as Kibana 9.5.4 could not run on Elasticsearch 8.19.23. It was solved by configuring them to have compatible versions.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Error Explanation**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of

concepts, etc.

Extent of Use: **Ai was used only to guide me fix the error.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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